ICT330

Timed Online Assignment - January Semester 2023

Database Management Systems

Monday, 15 May 2023 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **4** questions and comprises **7** pages (including cover page).

2. You must answer **4** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a Microsoft Word file saved in .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the Word file provided it does not exceed the file size limit of 500MB. **Images of handwritten answers will not be marked**.

6. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

(Full marks: 100)

Question 1

Tourists and locals alike read food reviews in various websites. It is helpful to have an application to collate reviews from different websites and to allow search by dish name (e.g., chicken rice), search by type of cuisine served (e.g., Korean), search by name of food establishment, search by price range of food establishment (in terms of the number of dollar or $ symbol), etc. The result of the search should show up food establishment(s), each with a general description, their outlet(s) and addresses, and web page links (urls) for reviews for the outlet(s). It is possible that there is currently no link for reviews for an outlet.

The names of food establishments are unique. The general description for each food establishment includes the start year of operation, the cuisine served, the price range, outlet address(es) and one to up to ten dishes with dish name and dish description which are specialties of the food establishment. The same specialty may be listed under more than one food establishment.

Each url leads to a page with food reviews for an outlet of a food establishment. You may assume that each review posted on a url has a unique timestamp. However, two reviews on two different urls can have the same timestamp. Each review includes a rating (in terms of number of stars) and may optionally include comments.

The application also keeps track of and displays the number of clicks a url has thus far.

Without adding any attribute/identifier not described in the question, construct a conceptual entity relationship (ER) model from the statements of requirements to represent the data model, showing

(i) Entities, with entities name, relevant attributes and identifier,
(10 marks)

(ii) Relationships, with maximum and minimum cardinalities and relationship name.
(10 marks)

State your assumption(s) for only data requirements that are not specified.

Question 2

A property agent wishes to set up a small database to manage his appointments. An appointment made is for one client to view one or more properties. A client may have multiple appointments, and a property may be viewed in multiple appointments.

If there are more than one viewer (e.g., a family) for an appointment, only one client details (contact and name) are recorded against the appointment. A client name is not unique, but client contact can identify a client.

You are given the table:

Viewing (propertyAddress, propertyType, askingPrice, clientName, clientContact, appointmentDatetime)

Do not introduce any new columns not in the given table. State your assumption(s) for only data requirements that are not specified.

Question 2a

Formulate and list the functional and multi-valued dependencies (MVD). Include at least one MVD that is subsumed and at least one MVD that is not subsumed.

(5 marks)

Question 2b

Draw dependency graph and propose candidate key(s) for the table. Show composite keys in brackets.

(3 marks)

Question 2c

(i) For each MVD in answer to Q2(a), explain why it is subsumed or not subsumed.

(ii) Normalise the table to Boyce Codd Normal Form (BCNF) and Fourth Normal Form (4NF) by applying **the BCNF approach** covered in the course text.

Show how you apply the steps to arrive at the tables, explain whether the structure of each derived table at each iteration is already in BCNF and 4NF (if applicable).

(10 marks)

Question 2d

Assemble your BCNF and 4NF tables and constraints clearly by giving each table and the referential integrity constraint(s), if any, in the forms described below:

Tables in this form:

TABLE-NAME(col-name$_1$, col-name$_2$, ..., col-name$_n$)

- indicate (PK) against the primary key and
- indicate (FK) against the foreign keys, if any.

Referential integrity constraint(s), if any, **together with the child table** in this form:

TABLE-NAME$_x$.col-name$_{x1}$ ...col-name$_{xn}$ must exist in

TABLE-NAME$_y$.col-name$_{y1}$...col-name$_{yn}$

(7 marks)

Question 3

Question 3a

(i) Transform the ERD in Figure Q3(a)(i) into a logical model.

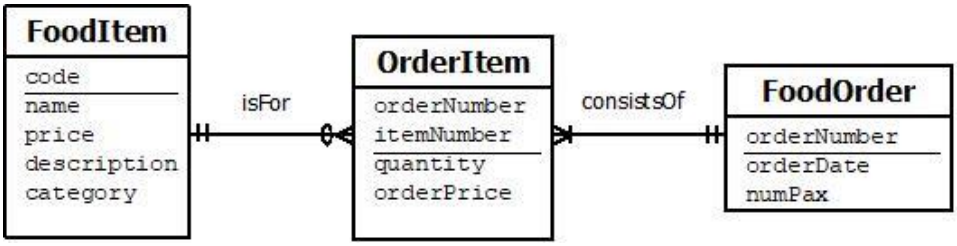

Figure Q3(a)(i)

(ii) Refer to the description of the columns for the column definition, and then code the appropriate SQL statements to implement ALL tables derived for the ERD in Figure Q3(a)(i).

Arrange your SQL statements in the order that they should be executed.

All columns are required except for the column description.

code	Consists of one alphabet followed by 2 digits.
name	At most 50 characters in length
price	A non-negative and non-zero decimal number
description	At most 256 characters in length
category	At most 50 characters in length, currently restricted to the values: main, soup, dessert, drink, salad
orderNumber	An integer, auto generated with starting value 1 and increment 1
itemNumber	An integer which is running for each order. In Q3(c), a trigger generates itemNumber for a FoodOrder.
quantity	An integer between 1 and 15 with default value 1
orderPrice	A decimal number. In Q3(c), a trigger assigns orderPrice to the current price of the FoodItem at the time of order
orderDate	A date with default today's date
numPax	A non-negative and non-zero integer with default value 1

Constraints for the relationships in Figure Q3(a)(i):

isFor	delete and update are disallowed
consistsOf	delete cascade and update disallowed

(10 marks)

Question 3b

Assume the following data rows are already in the database. You need NOT populate your tables as it will consume time. The data is for reference only for the queries you are to formulate.

	code	name	price	description	category	orderNumber	itemNumber	quantity	orderPrice	code	orderDate	numPax
1	A01	Braised Lamb Shank	28.90	Meltingly tender braised lamb shank	main	3	1	3	28.90	A01	2023-05-25	4
2	A01	Braised Lamb Shank	28.90	Meltingly tender braised lamb shank	main	5	1	3	28.90	A01	2023-05-26	5
3	A01	Braised Lamb Shank	28.90	Meltingly tender braised lamb shank	main	9	1	1	28.90	A01	2023-05-27	3
4	A02	Rosemary Lamb Chop	32.90	Special marinate lamb chop with herb and spices	main	2	1	3	32.90	A02	2023-05-25	4
5	A02	Rosemary Lamb Chop	32.90	Special marinate lamb chop with herb and spices	main	3	2	1	32.90	A02	2023-05-25	4
6	A02	Rosemary Lamb Chop	32.90	Special marinate lamb chop with herb and spices	main	5	2	3	32.90	A02	2023-05-26	5
7	A02	Rosemary Lamb Chop	32.90	Special marinate lamb chop with herb and spices	main	7	2	1	32.90	A02	2023-05-27	2
8	A02	Rosemary Lamb Chop	32.90	Special marinate lamb chop with herb and spices	main	8	1	1	32.90	A02	2023-05-27	1
9	A02	Rosemary Lamb Chop	32.90	Special marinate lamb chop with herb and spices	main	9	2	2	32.90	A02	2023-05-27	3
10	A03	Grilled Beef Steak	29.90	The most tender and juicy thick cut	main	1	1	3	29.90	A03	2023-05-25	3
11	A03	Grilled Beef Steak	29.90	The most tender and juicy thick cut	main	4	1	8	29.90	A03	2023-05-26	8
12	A03	Grilled Beef Steak	29.90	The most tender and juicy thick cut	main	6	1	1	29.90	A03	2023-05-27	1
13	A03	Grilled Beef Steak	29.90	The most tender and juicy thick cut	main	7	1	1	29.90	A03	2023-05-27	2
14	B01	Manhattan Clam Ch...	7.90	Cream Soup with tomato base served with fresh clam	soup	NULL	NULL	NULL	NULL	NU...	NULL	NULL
15	B02	Golden Seafood Ch...	8.90	Pumpkin Broth with Fresh Sauteed Seafood	soup	6	3	1	8.90	B02	2023-05-27	1
16	B03	Cream Of Mushroom...	6.90	Rich Flavour of fresh mushroom with truffle oil	soup	4	2	4	7.90	B03	2023-05-26	8
17	C01	Crazy Mud Pie	4.90	Our crazy dessert for fans	dessert	NULL	NULL	NULL	NULL	NU...	NULL	NULL
18	C02	Crispy Chocolate Wa...	3.90	Crispy waffle with vanilla ice-cream	dessert	4	7	1	3.90	C02	2023-05-26	8
19	C03	Matcha Lava Cake	3.90	Premium matcha and white chocolate served with ice-creams	dessert	4	6	7	3.90	C03	2023-05-26	8
20	D01	Wildberry Lychee Fr...	5.90	Jumbo Punch	drink	1	2	2	5.90	D01	2023-05-25	3
21	D01	Wildberry Lychee Fr...	5.90	Jumbo Punch	drink	3	5	2	5.90	D01	2023-05-25	4
22	D01	Wildberry Lychee Fr...	5.90	Jumbo Punch	drink	4	3	2	5.90	D01	2023-05-26	8
23	D01	Wildberry Lychee Fr...	5.90	Jumbo Punch	drink	6	2	1	5.90	D01	2023-05-27	1
24	D02	Peaches Pink	6.90	Signature Series	drink	2	3	4	6.90	D02	2023-05-25	4
25	D02	Peaches Pink	6.90	Signature Series	drink	3	4	2	6.90	D02	2023-05-25	4
26	D02	Peaches Pink	6.90	Signature Series	drink	4	4	6	6.90	D02	2023-05-26	8
27	D02	Peaches Pink	6.90	Signature Series	drink	5	4	4	6.90	D02	2023-05-26	5
28	D02	Peaches Pink	6.90	Signature Series	drink	7	3	2	6.90	D02	2023-05-27	2
29	D02	Peaches Pink	6.90	Signature Series	drink	8	3	1	6.90	D02	2023-05-27	1
30	D02	Peaches Pink	6.90	Signature Series	drink	9	4	3	6.90	D02	2023-05-27	3
31	D03	Hot Passion Fruits	4.90	Hot Beverage	drink	NULL	NULL	NULL	NULL	NU...	NULL	NULL
32	E01	Classic Caesar Salad	7.90	Traditional classic salad serve with our secret recipe dressing	salad	3	3	4	7.90	E01	2023-05-25	4
33	E01	Classic Caesar Salad	7.90	Traditional classic salad serve with our secret recipe dressing	salad	4	5	6	5.90	E01	2023-05-26	8
34	E01	Classic Caesar Salad	7.90	Traditional classic salad serve with our secret recipe dressing	salad	9	3	2	7.90	E01	2023-05-27	3
35	E02	Swiss Mushroom Sal...	9.90	Mixed salad tossed with tangy balsamic vinaigrette dressing with sauteed mushroom	salad	NULL	NULL	NULL	NULL	NU...	NULL	NULL
36	E03	Japanese Sesame ...	9.90	Cold pasta & shrimp infused with japanese dressing	salad	2	2	4	9.90	E03	2023-05-25	4
37	E03	Japanese Sesame ...	9.90	Cold pasta & shrimp infused with japanese dressing	salad	5	3	4	9.90	E03	2023-05-26	5
38	E03	Japanese Sesame ...	9.90	Cold pasta & shrimp infused with japanese dressing	salad	8	2	1	9.90	E03	2023-05-27	1

Formulate SQL queries to display the required resultsets:

(i) List food items that are not soup and cost less than $15. Sort the result by ascending order of category, price and name.

	code	name	price	category
1	C02	Crispy Chocolate Waffle	3.90	dessert
2	C03	Matcha Lava Cake	3.90	dessert
3	C01	Crazy Mud Pie	4.90	dessert
4	D03	Hot Passion Fruits	4.90	drink
5	D01	Wildberry Lychee Fruit Tea	5.90	drink
6	D02	Peaches Pink	6.90	drink
7	E01	Classic Caesar Salad	7.90	salad
8	E03	Japanese Sesame Shrimp with Cold Pasta	9.90	salad
9	E02	Swiss Mushroom Salad	9.90	salad

(ii) Food orders that do not have soup food items.

	orderNumber
1	1
2	2
3	3
4	5
5	7
6	8
7	9

(iii) The average cost per pax for each food order if the average cost per pax is above $40.

	orderNumber	numPax	Total Cost	Avg per Pax
1	2	4	165.90	41.475
2	3	4	176.80	44.20
3	4	8	390.60	48.825
4	5	5	252.60	50.52
5	6	1	44.70	44.70
6	8	1	49.70	49.70
7	9	3	131.20	43.7333

(15 marks)

Question 3c

Write a trigger to perform the following actions when an order item is added to a food order:

- If the order item is for a food item already in the order, do not add another order item but rather, add the quantity to existing quantity of the order item.
- If the order item is for a food item not yet in the order, generate the next item number, assign the order price as the current price of the food item, and add the order item to the order.

State any assumptions you made in implementing the trigger.

(10 marks)

Question 4

Question 4a

Demonstrate the basic techniques of concurrency control and their application.

A stored procedure A executes daily to compute interest earned to bank accounts based on their lowest balance for the day. Another stored procedure B executes to record deposit and withdrawal transactions made on the bank account. Propose the lowest level of isolation needed for stored procedures A and B. Explain your choice of isolation level.

(6 marks)

Question 4b

Describe **ONE (1)** permission that should be given to bank customers to allow them to deposit and withdraw money. Describe **ONE (1)** permission that should not be given to bank customers. Specify the permissions in terms of these actions: select, delete, insert and update on user tables in a bank setting.

(4 marks)

Question 4c

Given Figure 4(c), explain what the database recovery procedure will do for the transactions T1 to T3. Explain also how the transaction log help ensure the ACID properties of T1 to T3.

Record Pointer	Transaction ID	Reverse Pointer	Forward Pointer	Time	Type of Operation	Table	Row ID	Before Image	After Image
1	T1	0	2	14:01:32	Start				
2	T1	1	3	14:01:36	Insert	Table1	2		2, …
3	T1	2	7	14:01:55	Update	Table2	14	14, 2, …	14, 3, …
4	T2	0	5	14:02:01	Start				
5	T2	4	6	14:02:24	Insert	Table3	15		15, …
6	T2	5	10	14:03:41	Update	Table4	16	16, 7, …	16, 8, …
7	T1	3	0		Rollback				
Checkpoint									
8	T3	0	9	14:15:23	Start				
9	T3	8	12	14:05:00	Update	Table1	3	3, 5, …	3, 8, …
10	T2	6	11	12:44:02	Update	Table2	15	15, 2, …	15, 3, …
11	T2	10	0	12:44:25	Commit				
12	T3	9	0	14:43:33	Update	Table1	4	4, 7, …	4, 3, …
Crash!									

Figure 4(c)

(10 marks)

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